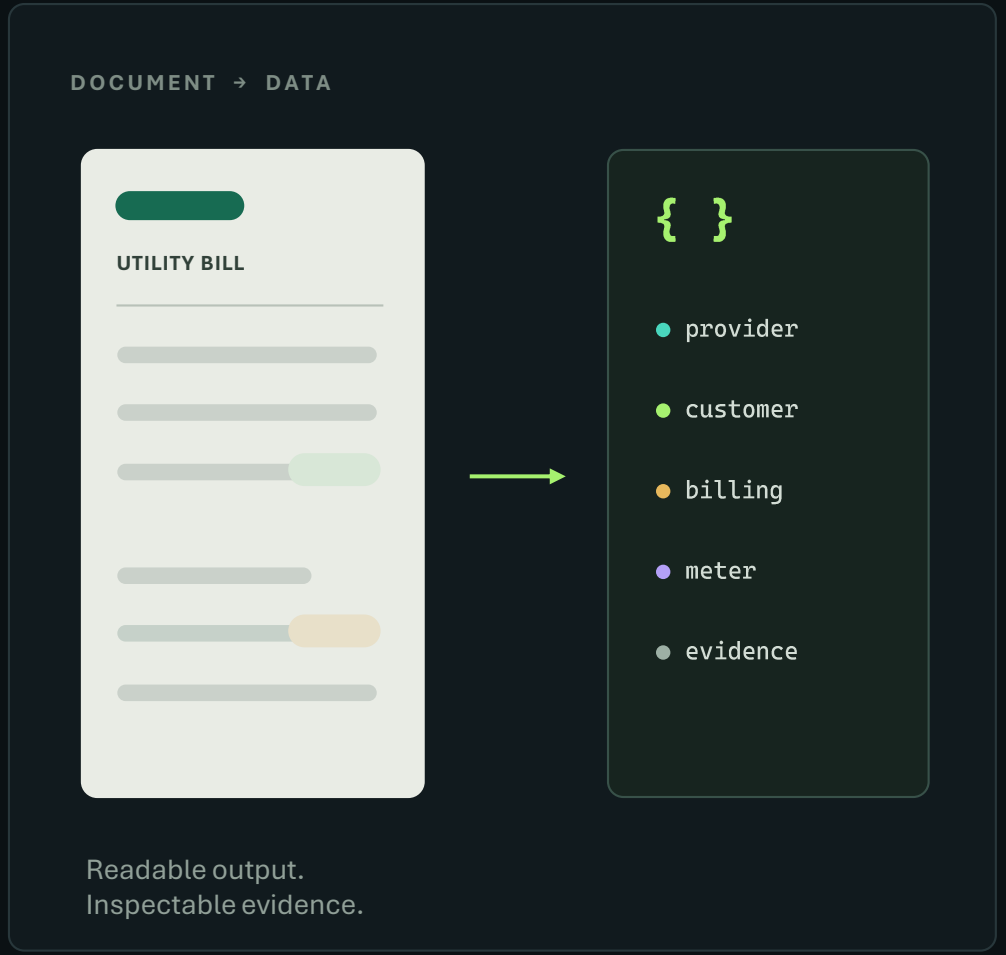
URJA

LOCAL-FIRST / DOCUMENT INTELLIGENCE

From messy utility documents to structured, validated data.

A document-understanding pipeline built for real Indian utility bills.

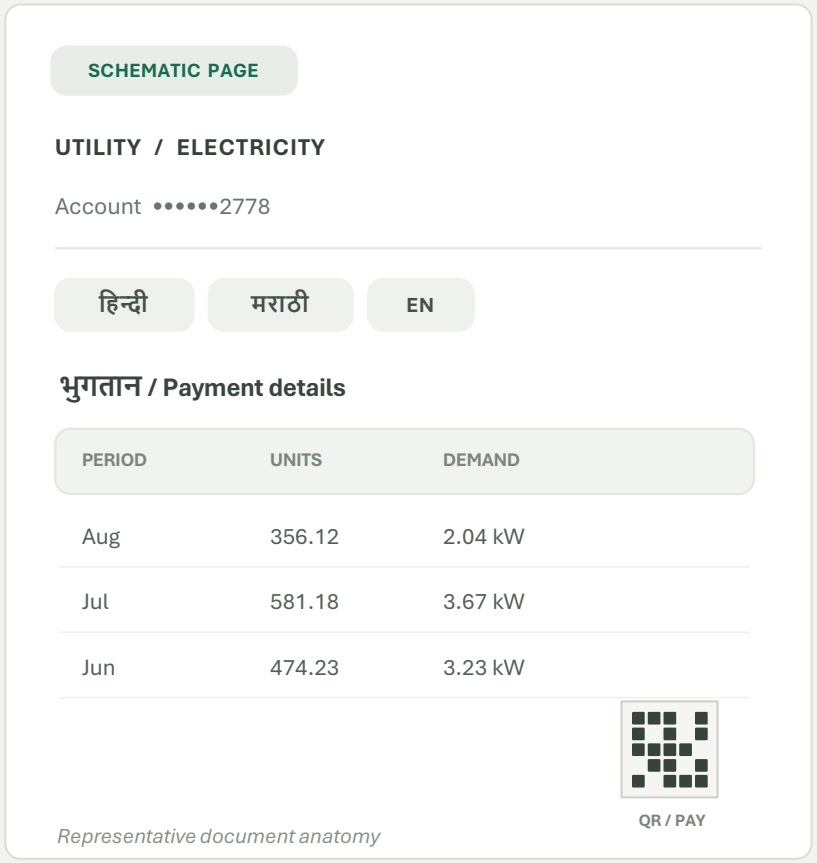
PDF + IMAGES | INDICOCR | STRUCTURED JSON



LOCAL PROCESSING · GENERIC DOCUMENT MODEL · DOMAIN-AWARE RESULTS

Utility bills are not just text

A single page can mix scripts, tables, identifiers, payment cues and provider-specific layouts.



01 LANGUAGE

Hindi, Marathi, English and mixed scripts

02 STRUCTURE

Tables, headings, notices and payment blocks

03 IDENTITY

Account references, customer details and dates

04 MEASUREMENTS

Meter readings, units and usage history

The challenge is not reading text.

The challenge is understanding the document.

Layout carries meaning: which value belongs to which account, period, table or payment action.

Plain OCR can read a value and still lose its meaning

Structure links text to rows, fields, pages and evidence.

TEXT-ONLY PATH

01 **PDF / PAGE IMAGE**
visual layout + multilingual text

↓

02 **PLAIN OCR TEXT**
tokens without table or field links

↓

03 **CONTEXT LOST**
values become hard to interpret

Text survives; relationships do not.

URJA STRUCTURED PATH

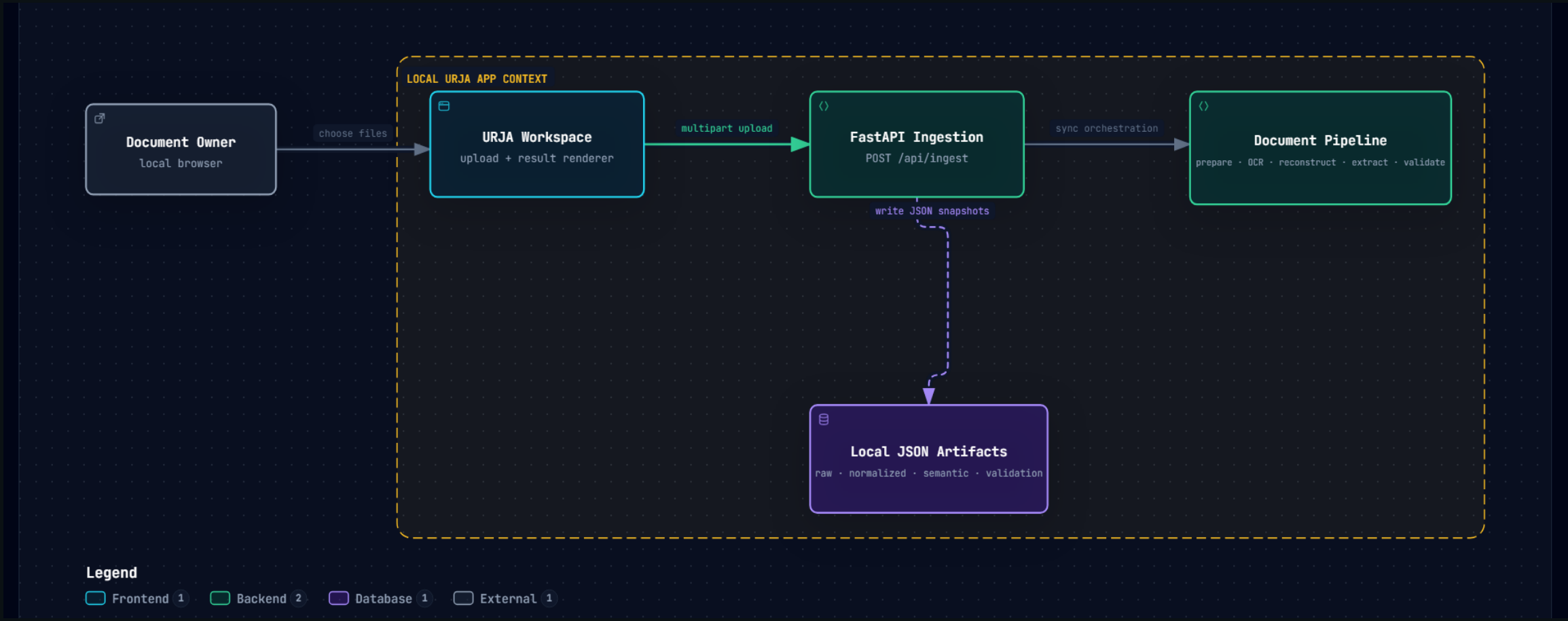
DOCUMENT	PDF / images
OCR	IndicOCR
LAYOUT	tables + sections
SEMANTICS	fields + evidence
CHECKS	validation

STRUCTURED JSON — fields / tables / source evidence

Interpretation depends on preserved relationships, not OCR alone.

A local processing path, with explicit file persistence

The API coordinates the current synchronous pass and writes inspectable JSON artifacts.

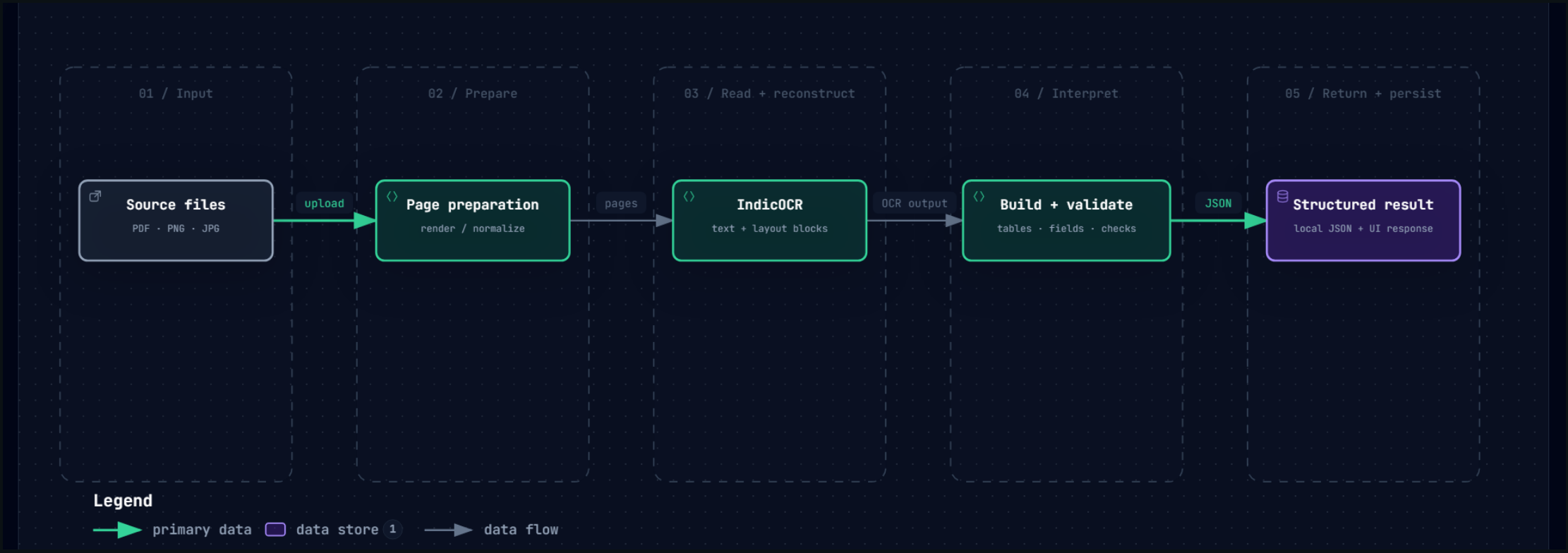


LOCAL APP CONTEXT

Filesystem output lives under data/documents/; the current API returns results after processing completes.

One ordered pass from uploaded files to dashboard data

PDFs become page images; the same per-page IndicOCR path handles uploaded images.

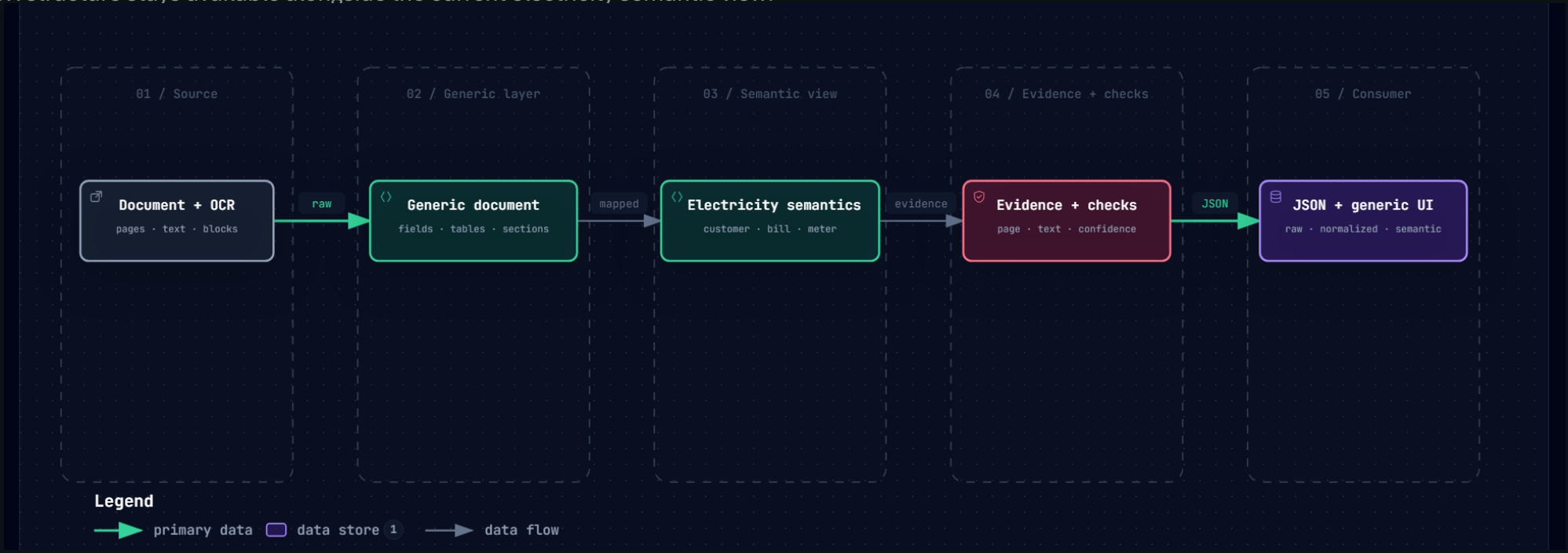


01 UPLOAD | 02 PREPARE | 03 OCR | 04 RECONSTRUCT | 05 EXTRACT | 06 VALIDATE | 07 SAVE JSON | 08 RENDER

Semantic extraction and deterministic checks currently specialize in electricity bills. Other types retain generic document structure.

A document record carries more than extracted values

Generic OCR structure stays available alongside the current electricity semantic view.



IDENTITY

provider · customer · account

BILLING

dates · amounts · payment

USAGE

meter · readings · history

DOCUMENT

tables · notes · notices

TRUST

validation · field evidence

Gas and water semantic adapters are future work; they are not current specialized extractors.

A four-page bill, reduced to reviewable information

Representative output from the URJA electricity-bill fixture; customer identifiers are masked.

SOURCE DOCUMENT

PURVANCHAL VIDYUT
VITARAN NIGAM LTD.

Account •••••2778

CURRENT BILL	₹2,414.98
PRIOR DUES	₹0.49
PAYABLE	**₹2,415.00**
METER READINGS	10,315.33 → 10,671.28
USAGE	355.95 kWh

4 pages / 11 OCR tables / 6 usage periods

TEST FIXTURE · MASKED IDENTITY

EXTRACTED FIELDS

PROVIDER
Purvanchal Vidyut Vitaran Nigam Ltd.

CUSTOMER
Anonymized fixture

ACCOUNT NUMBER
•••••2778

BILL DATE
02 Sep 2026

DUE DATE
17 Sep 2026

CURRENT BILL
₹2,414.98

PAYABLE
₹2,415.00

PREVIOUS DUES
₹0.49

METER READING
10,315.33 → 10,671.28

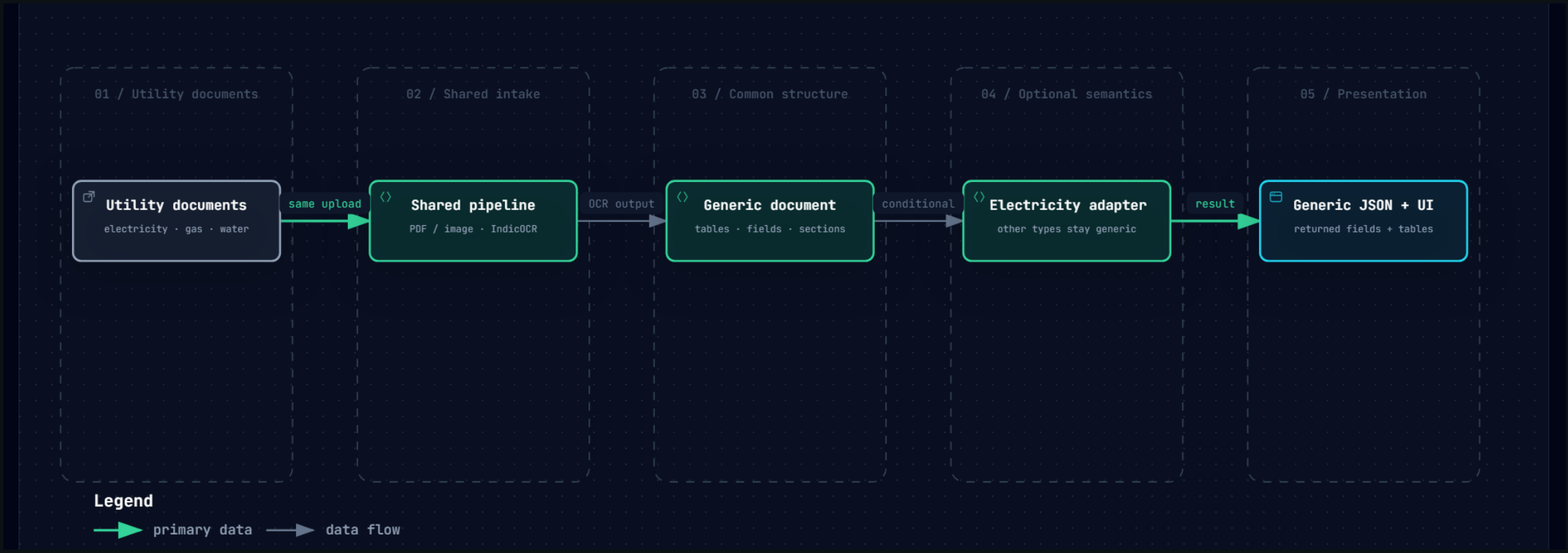
CONSUMPTION
355.95 kWh

3 VALIDATION CHECKS PASSED

Amounts and meter values shown are fixture outputs, not accuracy benchmarks.

One input path. Domain adapters grow by type.

Provider-specific layouts do not require provider-specific frontend pages.



CURRENT Generic document reconstruction + electricity semantics

PLANNED Gas / water domain mappings and broader provider coverage

Unsupported domains can retain generic OCR/document structure; specialized extraction is not implied.

A focused workspace, not an operations console

The browser shows a quiet indeterminate state while the current API performs its synchronous pass.

WORKING STATE

Processing your document

Sent to local processor

Processing locally

No live page or stage telemetry is exposed by the API.

No invented percentages or page counts.

INDETERMINATE BY DESIGN

DOCUMENT RESULTS

COMPLETED

test_bill.pdf

Electricity bill · 4 pages

PROVIDER	ACCOUNT	CURRENT BILL
Purvanchal Vidyut...	••••••2778	₹2,414.98

DUE DATE	USAGE	METER
17 Sep 2026	355.95 kWh	10,315.33 → 10,671.28

DETECTED TABLES | 11 OCR tables | 6 usage periods | Validation: passed

Identity values masked for presentation.

A working prototype, with a clear expansion path

Current scope is useful and inspectable; broader utility coverage still needs domain work.

IMPLEMENTED NOW

- ✓ PDF and image ingestion, with ordering and image grouping
- ✓ Per-page IndicOCR and explicit OCR failure status
- ✓ Generic page, field, table, section and visual representation
- ✓ Electricity semantic adapter with candidate evidence
- ✓ Deterministic validation and local JSON artifacts
- ✓ Frontend summary, generic tables and technical JSON views

NEXT

- → Semantic adapters for gas, water and more providers
- → Broader schema review for cross-utility common fields
- → Persisted document library and lifecycle management
- → Streaming processing-stage and page progress
- → More provenance review and conflict-resolution UX
- → Historical usage analysis and production hardening

NEXT MILESTONE Add the next domain adapter without changing the shared intake and rendering contract.